Lab Work 7: Data Transformation and Visualization with Power BI

Objective: Learn how to import financial institution data (from Excel and SQL Server) into Power BI, perform data transformations, and build visual reports.

Pre-requisites:

- **Excel file from Lab 4** (Financial Institution Data with columns: *Province, District, Class 'A', Class 'B', Class 'C', Total, Population, Pop. Per Branch*).
- **Lab 6 SQL Server summary tables** (aggregated results).
- **Power BI Desktop installed**.

Steps to Start:

1. **Download & Open Power BI Desktop.**
2. **Load Excel file from Lab 4**:
   - Go to **Home → Get Data → Excel → select the file**.
   - Load the sheet into Power BI.
3. **Transform Excel Data in Power Query** (click *Transform Data*):
   - Remove the `S.No.` column.
   - Rename columns (e.g., `Class 'A'` → `Class A`).
   - Ensure numeric columns (`Class A`, `Class B`, `Class C`, `Total`, `Population`, `Pop. Per Branch`) have the correct data type.
   - Create a **new calculated column**:
     - `Branches_Per_100K = (Total / Population) * 100000.`
   - Close & Apply.
4. **Connect Power BI to SQL Server database (Lab 6 results):**
   - Go to **Home → Get Data → SQL Server → enter server name and database**.
   - Select summary tables created in Lab 6.
   - Load them into the model.

5. Save your Power BI report file as `Lab7.pbix`.

Procedure

1. **Data Import:**
   - Imported the Excel file from Lab 4 into Power BI using *Home → Get Data → Excel*.
   - Selected the financial institution dataset and loaded it into the Power BI model.
2. **Data Transformation in Power Query:**
   - Removed unnecessary columns such as *S.No*.
   - Renamed fields for clarity (e.g., *Class 'A' → Class A*).
   - Ensured branch counts and population columns were properly set to numeric data types.
3. **SQL Server Connection:**
   - Connected Power BI to the SQL Server database containing summary tables created in Lab 6.
   - Imported the aggregated results (e.g., province-level totals and averages).
4. **Merging Data Sources:**
   - Integrated Excel-based branch data with SQL Server summary data.
   - Verified consistency across both sources for analysis.
5. **Visualization Development:**
   - Created a **Bar Chart** to show total branches by province.

- Built a **Pie Chart** to display Class A, B, and C distribution for each province.
- Used a **Line Chart** to show population-per-branch trends across provinces.
- Plotted a **Map Chart** showing district-level branch counts.
- Designed a **Stacked Column Chart** to compare total branches against population by province.

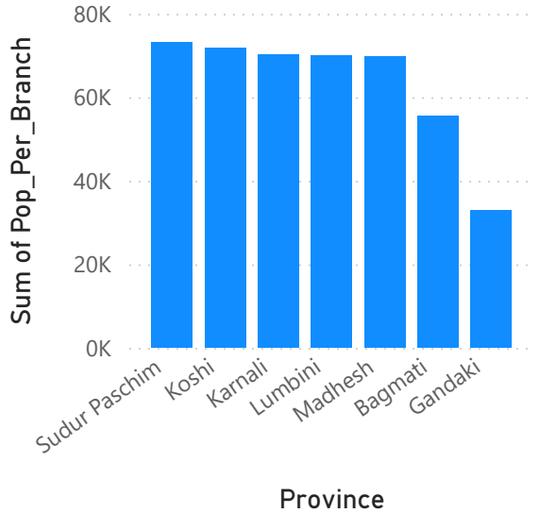
Tasks

1. **Bar Chart** – Show **Total Branches by Province**.
2. **Pie Chart** – Show **distribution of Class A, Class B, and Class C branches** across a selected province.
3. **Line Chart** – Show **Population per Branch** trend by province.
4. **Map Chart** – Show **Districts and their Total Branches** on a map.
5. **Stacked Column Chart** – Compare **Total Branches vs. Population** across provinces.

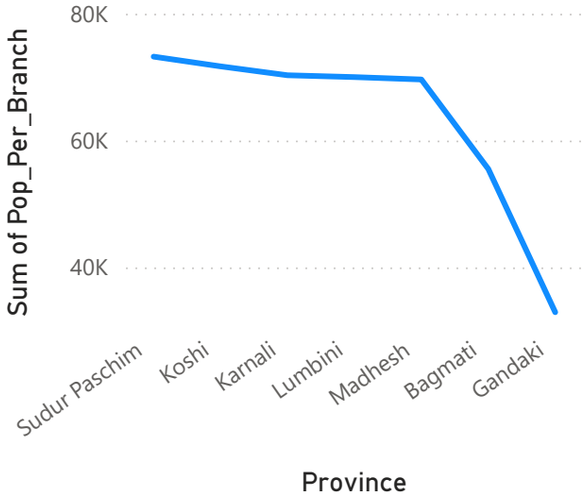
Additional Learning Outcomes:

- Practice **cleaning & transforming data** in Power Query.
- Learn to **merge Excel and SQL data** inside Power BI.
- Create **five different types of charts** for financial institution data.
- Understand how **visuals support decision-making** (e.g., identifying underserved provinces).

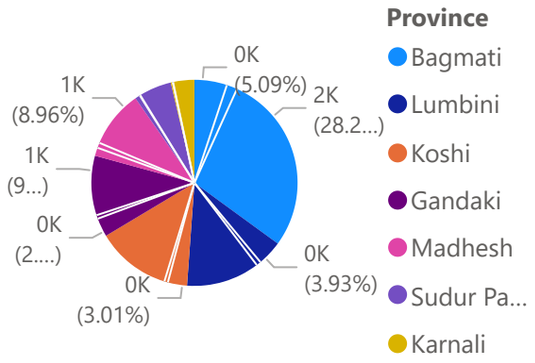
Sum of Pop_Per_Branch by Province



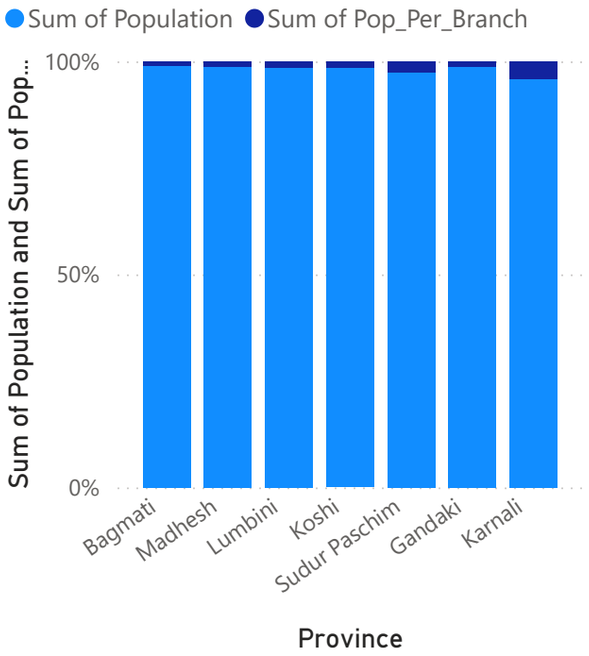
Sum of Pop_Per_Branch by Province



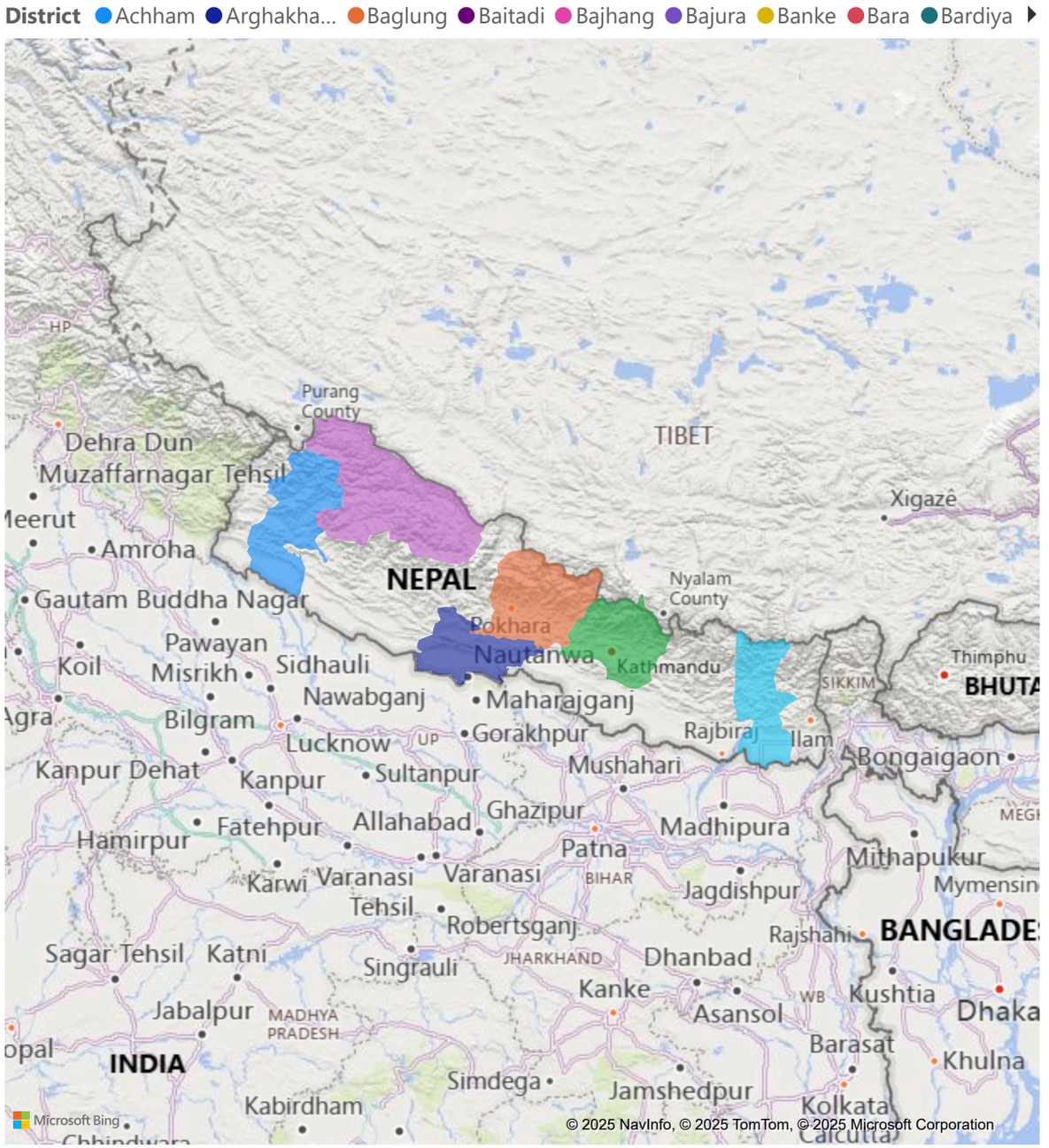
Sum of Class_B, Sum of Class_C and Sum of Class_A by Province



Sum of Population and Sum of Pop_Per_Branch by Province



First Latitude, First Longitude and Sum of Pop_Per_Branch by Province and District



Interpretation

1. Bar Chart → Total Branches by Province

- Provinces like **Bagmati** and **Lumbini** have the **highest number of total branches**, showing financial concentration in central and more urbanized regions.
- **Karnali** and **Sudur Paschim** have **significantly fewer branches**, indicating limited financial access in those provinces.
- The overall distribution suggests **banking services are not evenly spread**, favoring developed provinces.

2. Pie Chart → Distribution of Class A, B, C by Province

- **Class A banks (large commercial banks)** dominate in **Bagmati**, showing centralization of major financial institutions in the capital region.
- **Class B (development banks)** and **Class C (finance companies)** are more evenly spread, but still fewer in remote provinces.
- This highlights how **major banking resources are clustered**, while rural provinces rely on smaller financial institutions.

3. Line Chart → Population per Branch Trend

- Provinces like **Bagmati** have **lower population per branch**, meaning **better service availability** (more branches for fewer people).
- In contrast, **Madhesh, Karnali, and Sudur Paschim** show **high population per branch ratios**, suggesting **overloaded branches** and underserved populations.
- The trend line emphasizes **regional imbalance in financial access**.

4. Map Chart → District-Level Branch Counts

- District-level mapping shows **urban districts (e.g., Kathmandu, Lalitpur, Pokhara)** having **dense branch coverage**.
- Remote districts in **Karnali and Sudur Paschim** have **very few or scattered branches**.
- This geographic visualization clearly points out **underserved rural areas** and can guide policymakers on where to expand banking services.

5. Stacked Column Chart → Compare Total Branches vs Population

- **Bagmati and Madhesh**: High population **and** high total branches, but still a **mismatch when normalized** (population growing faster than branches).
- **Karnali & Sudur Paschim**: Low population **and** very low total branches → still underserved.
- **Lumbini & Koshi**: Show more balanced ratios between population and branch count.
- This chart makes it clear that **branch distribution does not scale proportionally with population size**.

Overall Interpretation: Your visualizations clearly show that **Nepal's banking services are concentrated in urban and central provinces (Bagmati, Lumbini, Koshi), while remote provinces (Karnali, Sudur Paschim) remain underserved**. Population vs. branch analysis highlights **inequality in financial access**, and class-wise distribution shows **large banks preferring central hubs** while smaller banks cover other areas.

Conclusion

In this lab, Power BI was used to transform, integrate, and visualize financial institution data from both Excel and SQL Server. The process involved cleaning raw data, renaming and standardizing fields, creating calculated measures, and combining multiple data sources for deeper insights. By building diverse visualizations, the analysis revealed variations in branch distribution across provinces and highlighted underserved areas when normalized by population.

This lab demonstrated how Power BI supports data-driven decision-making by converting raw tabular data into interactive, meaningful reports. Skills gained include data preparation with Power Query, creating calculated columns in DAX, integrating multiple data sources, and designing effective visual dashboards for analysis.